

Phase and Frequency Encoding

Biophysics 03-230

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1 Background

1.1 Larmor Equation Review

The frequency at which magnetization precesses after being tipped into the transverse plane is given by $\omega = \gamma B$, where γ is a nucleus-dependent gyromagnetic ratio constant in units of rad/s/Tesla and B is the static magnetic field strength in Tesla. *A higher magnetic field will increase the frequency while a lower magnetic field decreases frequency.* At the nominal field strength of the scanner $\mathbf{B}_0$, we define the Larmor frequency as $\omega_0 = \gamma \mathbf{B}_0$. For example, at a field strength of 3.0 T, ^1H nuclei will resonate at 128 MHz – just at the upper end of radio frequencies that carry your favorite FM stations.

1.2 Magnetization Phase

After magnetization is tipped into the transverse plane, it is complex-valued (i.e., it can be described as $M = M_x + iM_y$ with $i = \sqrt{-1}$). We will define the magnetization immediately after the RF pulse to be given by M^+ . If the magnetization is resonating at a frequency of $\Delta\omega$, the magnetization as a function of time t after the RF pulse will be given by $M(t) = M^+ \cos(\Delta\omega t) + iM^+ \sin(\Delta\omega t)$. More compactly, we can use the following nomenclature: $M(t) = M^+ e^{i\Delta\omega t}$. *Phase describes the angle by which the magnetization in the transverse plane has rotated around the field B_0 after excitation.*

1.3 Gradient Field Review

The x , y , and z gradient coils can be used to linearly change the B_0 field (pointing along the z dimension) as a function of position along each axis as:

$$B_0(x, y, z) = \mathbf{B}_0 + G_x x + G_y y + G_z z \quad (1)$$

Here, $G_{x,y,z}$ specifies the strength of the gradient are in units of Tesla/m. Maximum values of gradient strength achievable on whole-body human scanners are typically around 0.04-0.05 T/m.

Since the signals that we measure in MRI are demodulated (or recorded relative to) ω_0 , we can omit the $\mathbf{B}_0$ term containing the static field strength and consider only the contributions of the gradients:

$$B_0(x, y, z) = G_x x + G_y y + G_z z \quad (2)$$

1.4 Manipulating Phase Using Gradient Fields

If we apply a gradient along the x axis, we can see that $B_0(x) = G_x x$. Plugging this into the Larmor equation shows that the frequency of the magnetization is dependent on the position along the x axis: $\omega(x) = \gamma G_x x$.

We can also manipulate the phase of the magnetization as a function of x by turning on the G_x gradient only for a certain amount of time T . The phase, in units of radians, will be given by:

$$\phi(x) = \gamma G_x x T \quad (3)$$

More generally, the amplitude of G_x can be time-varying, and the phase will be given by the following integral.

$$\phi(x) = \gamma \int_0^T G_x(t) x dt \quad (4)$$

This shows us that the area underneath a time-varying gradient is what ultimately determines the phase of magnetization as a function of x .

1.5 Fourier Transforms of Time-Varying Signals

A periodic signal can be approximated with an infinite sum of sine and cosine functions of different frequencies. The Fourier transform is a way to calculate the weight given to the sine and cosine waves at each frequency such that when you add them all up, they approximate the original signal. Adding more and more sine and cosine waves will make the approximation more and more accurate.

Lets say we have a time-dependent signal $m(t)$, with time t being in units of seconds. We want to calculate the amplitudes of sinusoids at frequencies f (in units of 1/s or Hz) such that we can approximate the original signal. This is given by the following relationship:

$$M(f) = \int_{-\infty}^{+\infty} m(t) e^{i2\pi f t} dt \quad (5)$$

2 Spatial Encoding in One Dimension: Using Gradient Fields to Build a Fourier Relationship

The goal of "spatial encoding" in MRI is to *use the gradient coils to reverse engineer Equation 5 so that we can use an inverse Fourier transform to reconstruct the image*. Our signal (image) of interest is now $m(y)$, which can be thought of the "proton density" that varies along the y -axis.

Similar to the frequency values of the sinusoids used to represent a time-varying signal in Section 1.5, we will introduce a *spatial frequency* variable that will be used to reconstruct images that vary in the y dimension. The spatial frequency variable is referred to as a k -space coordinate and is given in units of 1/m (which is like the 1/s or Hz units used for f for time-varying signals).

So, re-writing Equation 5 in the desired terms ($t \rightarrow y$ and $f \rightarrow k_y$) gives:

$$M(k_y) = \int_{-\infty}^{+\infty} m(y) e^{i2\pi k_y y} dy \quad (6)$$

Now we want to define a relationship that will tell us how to use the time-varying $G_y(t)$ gradient to introduce the phase across the y dimension for each step n . Setting the phasor term in Equation 6 equal to our equation for the phase accrued by switching on a time-dependent gradient $G_y(t)$ (Equation 4) gives us the following:

$$2\pi k_y y = \gamma \int_0^T G_y(t) y dt \quad (7)$$

y can be taken outside of the integral since it is not dependent on t and each side of the equation can be divided by y to yield:

$$k_y = \frac{\gamma}{2\pi} \int_0^T G_y(t) dt \quad (8)$$

MRI data are measured as a function of k , and the inverse Fourier transform is used to reconstruct the underlying image of interest. However, we do not deal with continuous sampling as seen above. In reality, we measure discrete samples in k -space leading to a discrete image.

This discrete Fourier transform relationship is given by the following:

$$M(n\Delta k) = \sum_{j=-N/2}^{N/2-1} m_j e^{i2\pi n\Delta k y_j} \quad (9)$$

Here, the so-called step size in k -space is related to the imaging field-of-view (FOV) by $\Delta k = 1/\text{FOV}$, and the integer $n \in [-N/2, -N/2 + 1, \dots, N/2 - 1]$ controls the discrete values of spatial frequencies at which we will be measuring the MRI signal induced in the RF coil.

As we went over in class, the index n can be thought of as introducing a sinusoidal phase with precisely n periods repeated across the FOV.

3 Phase Encoding

In MRI, the term "phase encoding" refers to the repetition of RF pulses, gradient pulses, and data acquisition periods while changing the index n . In practice, a time-dependent gradient waveform will be calculated to get to the very edge of k -space as quickly as possible (i.e., to a value of $\frac{N\Delta k}{2}$, where N is the number of desired pixels in the image). The amplitude of the gradient waveform is then simply scaled down to sample different spatial frequencies (i.e., different values of n). This can be seen in Figure 1.

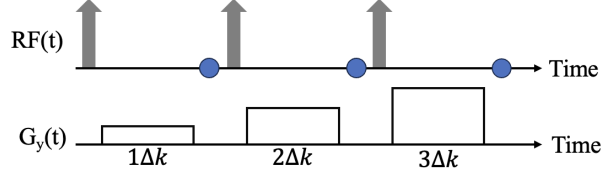


Figure 1: A basic sequence of RF and gradient pulses used to perform phase encoding. Immediately after each RF excitation, a phase encoding gradient is played out to introduce a phase across the FOV corresponding to the desired k -space coordinate. The voltage induced in the RF coils from the phase-modulated magnetization is then measured after the pulse is complete as indicated by the blue circles. This process is repeated to measure all desired values of the MRI signal at different coordinates $n\Delta k$.

3.1 Example: Designing Phase Encoding Gradient Pulses

Since it is not feasible to change the current in a gradient coil instantaneously, we have to ramp up our gradient amplitude, keep it at a maximum value for a certain period of time, and then ramp it back down. The area underneath this trapezoidal gradient waveform will then be related to our k -space coordinate. The goal of this example is to calculate the minimum amount of time it takes to get to the furthest point out in k -space that we need to acquire. This is important for making sure scan times are as short as possible.

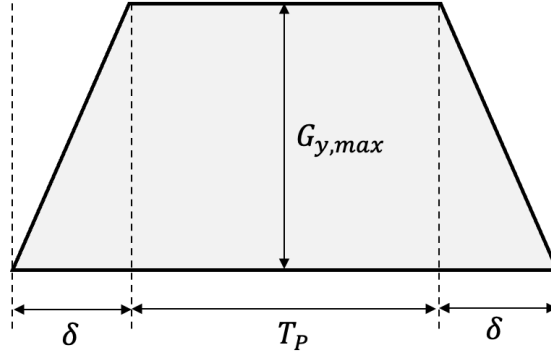


Figure 2: Trapezoidal $G_y(t)$ gradient pulse. The waveform starts at a value of zero and ramps up to the maximum value, stays at the maximum value for a period of time, and then ramps back down to zero.

Let's start with the following parameters:

- $\text{FOV} = 0.2 \text{ m}$
- $N = 256$
- $G_{y,\text{max}} = 30 \text{ mT/m}$
- $(dG_y/dt)_{\text{max}} = 200 \text{ T/m/s}$
- $\frac{\gamma}{2\pi} = 42.58 \times 10^6 \text{ Hz/T}$

3.1.1 How Far Out in k -Space Do We Need to Go?

We know that $\Delta k_y = 1/\text{FOV} = 1/0.2 = 5 \text{ m}^{-1}$. Since $N = 256$, we will need to go as far out as $N\Delta k/2$, or $k_{y,\text{max}} = \frac{256}{2}5 \text{ m}^{-1} = 640 \text{ m}^{-1}$.

3.1.2 Solve for Gradient Ramp-Up/Down Times

Using the maximum constraint on both the gradient amplitude itself and the slew rate (how fast the gradient can change), we will calculate the time it takes (δ) to ramp up the gradient to its maximum value.

$$\delta = \frac{G_{y,\text{max}}}{(dG_y/dt)_{\text{max}}} = \frac{30 \times 10^{-3} \text{ T/m}}{200 \text{ T/m/s}} = 0.00015 \text{ s} = 0.15 \text{ ms} \quad (10)$$

3.1.3 Calculate Duration of Gradient Plateau

First we need to calculate how far in k -space we have gone just from ramping up/down the gradients.

$$k_{y,\text{ramp}} = 2 \frac{\gamma}{2\pi} \left(\frac{1}{2} \delta G_{y,\text{max}} \right) = (42.58 \times 10^6 \text{ Hz/T})(0.00015 \text{ s})(0.03 \text{ T/m}) = 191.61 \text{ m}^{-1} \quad (11)$$

Note that this is just the equation for a triangle and the factor of 2 comes from the ramp-up and the ramp-down of the gradient.

Now we calculate the duration of the gradient plateau (T_P) in order to achieve the difference between the maximum desired and the already-achieved k -space coordinates.

$$(k_{y,\text{max}} - k_{y,\text{ramp}}) = \frac{\gamma}{2\pi} G_{y,\text{max}} T_P \quad (12)$$

$$T_P = \frac{640 - 191.61 \text{ m}^{-1}}{(42.58 \times 10^6 \text{ Hz/T})(0.03 \text{ T/m})} \approx 0.000351 \text{ s} \quad (13)$$

The total duration of the gradient pulse is then given by $2\delta + T_P = 2 \cdot (0.15 \text{ ms}) + 0.351 \text{ ms} = 0.651 \text{ ms}$.

4 Frequency Encoding

While phase encoding is a very effective method for performing spatial encoding, it is very slow; An excitation pulse and repetition time (TR) period is needed to collect every single k -space sample.

If we need to acquire a 3D image at a high spatial resolution, scan time will increase exponentially. *Frequency encoding is a much faster way to collect k -space samples for spatial encoding.* All k -space samples along the frequency encoding dimension are acquired after a single excitation RF pulse.

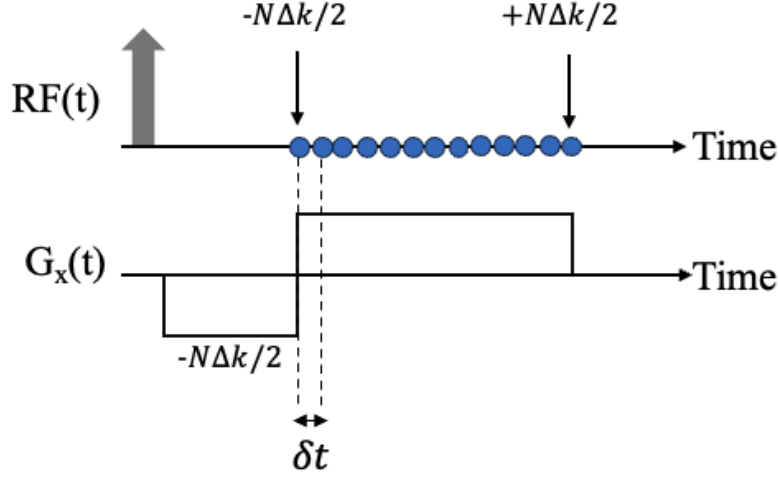


Figure 3: In frequency encoding, all k -space samples are acquired very quickly *while a gradient pulse is switched on*, leading to a very efficient acquisition.

On the scanner user interface, the technologist or researcher will choose the "readout bandwidth" (or "BW" in Hz), which determines the time spacing between the frequency encoding samples with the following relationship: $\delta t = 1/\text{BW}$. The logic in the pulse sequence program must then calculate the amplitude of the G_x gradient to make sure that exactly $1\Delta k$ is traversed in a time δt . This is done with the following equation:

$$G_x = \frac{2\pi\Delta k}{\gamma\delta t} \quad (14)$$

4.1 Frequency Encoding Artifacts from Imperfections in the Magnetic Field

As seen in Figure 3, each k -space sample is acquired at a different point in time relative to the excitation pulse, so magnetization that is precessing slower or faster due to perturbations in the B_0 will continue accruing phase throughout the data sampling period.

$$M(n\Delta k) = \sum_{j=-N/2}^{N/2-1} m_j e^{i2\pi n\Delta k y_j} e^{i\gamma\Delta B_0 n\delta t} \quad (15)$$

Factoring out $2\pi n\Delta k$ from the second exponential term (using Equation 14) above leads to:

$$M(n\Delta k) = \sum_{j=-N/2}^{N/2-1} m_j e^{i2\pi n\Delta k \left(y_j + \frac{\Delta B_0}{G_x} \right)} \quad (16)$$

This tells us that pixels located at location y get shifted to a position $y + \frac{\Delta B_0}{G_x}$.

4.1.1 Example: Frequency Encoding Distortions

Given the following information, calculate how far each pixel is shifted along the frequency encoding axis:

- FOV = 0.2 m
- BW = 25 kHz
- $\frac{\gamma}{2\pi} = 42.58 \times 10^6$ Hz/T
- $\Delta B_0 = 10 \mu\text{T}$

First we calculate Δk : $\Delta k = 1/\text{FOV} = 1/0.2 = 5 \text{ m}^{-1}$

Next we can calculate the frequency encoding gradient amplitude:

$$G_x = \frac{2\pi\Delta k}{\gamma\delta t} = \frac{2\pi\Delta k \cdot \text{BW}}{\gamma} = \frac{(5 \text{ m}^{-1})(25,000 \text{ Hz})}{42.58 \times 10^6 \text{ Hz/T}} = 2.94 \text{ mT/m} \quad (17)$$

Now we can finally calculate how far each pixel shifts due to ΔB_0 :

$$\Delta y = \frac{10 \times 10^{-6} \text{ T}}{0.00294 \text{ T/m}} = 0.0034 \text{ m} = 3.4 \text{ mm} \quad (18)$$

5 A Complete 2D Pulse Sequence

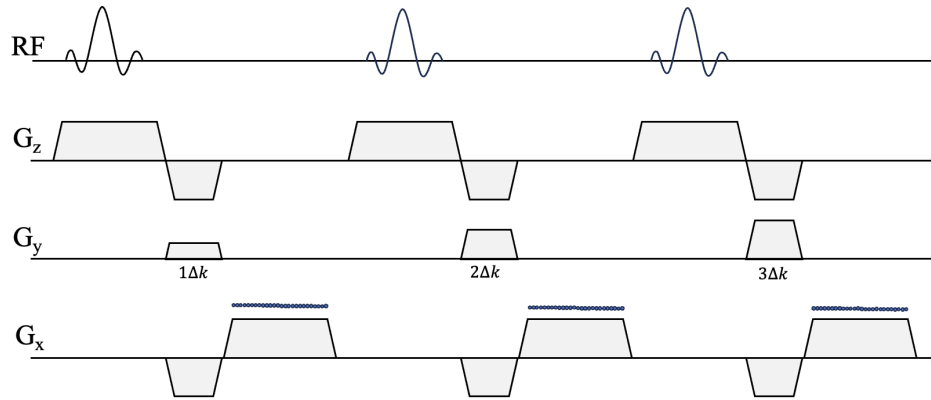
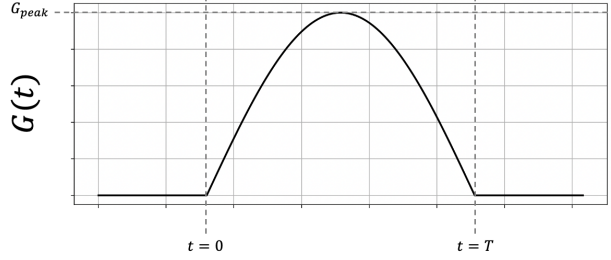


Figure 4: In a two dimensional pulse sequence, one of the two dimensions is phase encoded and the other is frequency encoded. So the slice-selective RF pulse is repeated and frequency encoding is performed for each desired phase encoding line.

1 Homework Problem 1

A gradient pulse that has a sinusoidal shape which varies in time as $G(t) = G_{\text{peak}} \sin\left(\frac{\pi t}{T}\right)$ is to be used for phase encoding. The pulse is depicted below.



1.1 Derive an expression for the k -space coordinate at time $t = T$.

$$k(T) = \frac{\gamma}{2\pi} \int_0^T G_{\text{peak}} \sin\left(\frac{\pi t}{T}\right) dt \quad (1)$$

$$k(T) = \frac{\gamma}{2\pi} G_{\text{peak}} \left[\frac{-T}{\pi} \cos\left(\frac{\pi T}{T}\right) - \frac{-T}{\pi} \cos\left(\frac{\pi \cdot 0}{T}\right) \right] \quad (2)$$

$$k(T) = \frac{\gamma}{2\pi} G_{\text{peak}} \left[\frac{-T}{\pi} \cos(\pi) + \frac{T}{\pi} \cos(0) \right] \quad (3)$$

$$k(T) = \frac{\gamma}{2\pi} G_{\text{peak}} \left[\frac{-T}{\pi} \cos(\pi) + \frac{T}{\pi} \cos(0) \right] \quad (4)$$

$$k(T) = \frac{\gamma}{2\pi} G_{\text{peak}} \frac{2T}{\pi} = \frac{\gamma G_{\text{peak}} T}{\pi^2} \quad (5)$$

1.2 Derive an expression for the rate change in the gradient pulse as a function of time. At what point in time is this function at its maximum? Evaluate this expression at this time.

$$\frac{dG}{dt} = \frac{d}{dt} \left[G_{\text{peak}} \sin\left(\frac{\pi t}{T}\right) \right] \quad (6)$$

$$\frac{dG}{dt} = \frac{G_{\text{peak}} \pi}{T} \cos\left(\frac{\pi t}{T}\right) \quad (7)$$

This function reaches its maximum value when $t = 0$, resulting in the following expression for the maximum rate in change in the gradient amplitude:

$$\left(\frac{dG}{dt}\right)_{\max} = \frac{G_{\text{peak}}\pi}{T} \quad (8)$$

- 1.3 Assuming $G_{\text{peak}} = G_{\text{max}} = 25 \text{ mT/m}$, what is the minimum duration T of the pulse that keeps the rate in change in the gradient amplitude at the hardware-defined maximum value of 100 T/m/s ?**

$$\left(\frac{dG}{dt}\right)_{\max} = \frac{G_{\text{peak}}\pi}{T} = 100 \quad (9)$$

$$T = \frac{25 \times 10^{-3} \text{ T/m} \cdot \pi}{100 \text{ T/m/s}} = 0.000785 \text{ s} = 0.785 \text{ ms} \quad (10)$$

- 1.4 Using your answers from 1.1 and 1.3, and assuming $\frac{\gamma}{2\pi} = 42.58 \text{ MHz/T}$, calculate the k -space coordinate at the end of the pulse.**

$$k(T) = \frac{\gamma G_{\text{peak}} T}{\pi^2} = 2 \frac{\gamma}{2\pi} \frac{G_{\text{peak}} T}{\pi} \quad (11)$$

$$k(T) = \frac{2 \cdot (42.58 \times 10^6 \text{ Hz/T})(25 \times 10^{-3} \text{ T/m})(0.000785 \text{ s})}{\pi} = 531.98 \text{ m}^{-1} \quad (12)$$

- 1.5 Calculate the desired maximum k -space coordinate assuming a field-of-view of 0.25 m and an imaging matrix size of 256 .**

$$k_{\max} = \frac{N\Delta k}{2} = \frac{N}{2FOV} = \frac{128}{0.25 \text{ m}} = 512 \text{ m}^{-1} \quad (13)$$

- 1.6 Does the area of the pulse need to be increased or decreased to get to $k_{\max}$? Would you recommend adjusting the gradient amplitude or the duration of the pulse to achieve $k_{\max}$? Explain your answer and calculate the final values for the gradient pulse amplitude and duration accordingly (i.e., G_{final} and T_{final}).**

The area must be decreased to achieve $k_{\max}$. Since the duration was already calculated as to push the rate change of the gradient to its limit, the pulse cannot be shortened any more without exceeding the hardware limitation. Therefore, the gradient pulse amplitude will be reduced.

$$G_{\text{final}} = \frac{k_{\text{max}}}{k(T)} G_{\text{peak}} = \frac{512 \text{ m}^{-1}}{531.98 \text{ m}^{-1}} (0.025 \text{ T/m}) = 24.1 \text{ mT/m} \quad (14)$$

$$T_{\text{final}} = 0.785 \text{ ms} \quad (15)$$

1.7 Without changing the duration, what must the amplitude of the gradient pulse be for the phase encoding line at a k -space coordinate of $-5\Delta k$?

$$G_{-5\Delta k} = \frac{-5}{\text{FOV} \cdot k_{\text{max}}} G_{\text{max}} = \frac{-5 \cdot (0.025 \text{ T/m})}{(0.25 \text{ m})(512 \text{ m}^{-1})} = -0.000977 \text{ T/m} = -0.977 \text{ mT/m} \quad (16)$$

2 Homework Problem 2

2.1 Assuming a field-of-view of 0.25 m, magnetic field error of $20\mu\text{T}$, and $\frac{\gamma}{2\pi} = 42.58 \text{ MHz/T}$, calculate the minimum bandwidth needed to keep frequency encoding shifts less than $\Delta x = 1 \text{ mm}$.

$$\frac{\Delta B_0}{G} < \Delta x \quad (17)$$

$$\frac{\gamma \Delta B_0}{2\pi \Delta k \cdot \text{BW}} < \Delta x \quad (18)$$

$$\text{BW} > \frac{\gamma \Delta B_0}{2\pi \Delta k \Delta x} \quad (19)$$

$$\text{BW} > \frac{\text{FOV} \cdot \gamma \Delta B_0}{2\pi \Delta x} \quad (20)$$

$$\text{BW} > \frac{(42.58 \times 10^6 \text{ Hz/T}) \cdot (20 \times 10^{-6} \text{ T}) \cdot (0.25 \text{ m})}{0.001 \text{ m}} \quad (21)$$

$$\text{BW} > 212,900 \text{ Hz} \quad (22)$$